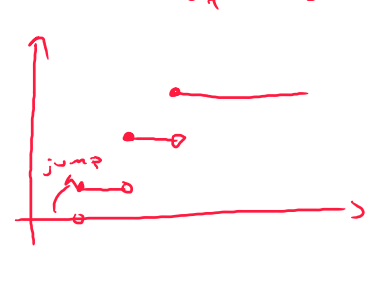


Mixed Random Variables

CDF of a Discrete R.V.



CDF of a Continuous R.V.



CDF of mixed R.V.



EX/ X_i C.R.V.

$$f_X(x) = \begin{cases} 2x, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

$$Y = g(X) = \begin{cases} X, & 0 \leq X \leq \frac{1}{2} \\ \frac{1}{2}, & X > \frac{1}{2} \end{cases} \quad \text{Find CDF of } Y$$

solution $R_X = [0, 1]$

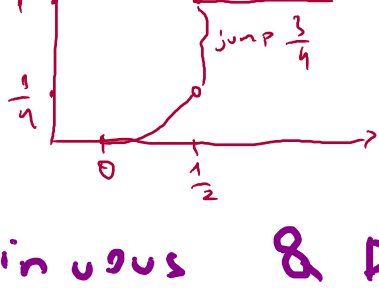
$$R_Y = \{y(x) | x \in R_X\} = [0, \frac{1}{2}]$$

$$F_Y(y) = \begin{cases} 0, & y < 0 \\ \frac{1}{3}, & 0 \leq y \leq \frac{1}{2} \\ 1, & y > \frac{1}{2} \end{cases}$$

$$P(Y = \frac{1}{2}) = P(X \geq \frac{1}{2}) = \int_{\frac{1}{2}}^1 2x dx = \frac{1}{2}$$

$$F_Y(y) = P(Y \leq y) = P(X \leq y) = \int_0^y 2x dx = y^2$$

Graph



Y is a mixed R.V.

Continuous & Discrete PARTS of M.R.V.

Y: Mixed R.V.

$$F_Y(y) = \underbrace{C(y)}_{\text{Continuous Part}} + \underbrace{D(y)}_{\text{Discrete Part}}$$

EX continued

$$C(y) = \begin{cases} 0, & y < 0 \\ y^2, & 0 \leq y < \frac{1}{2} \\ \frac{1}{3}, & y \geq \frac{1}{2} \end{cases} \quad D(y) = \begin{cases} \frac{1}{3}, & y \geq \frac{1}{2} \\ 0, & \text{otherwise} \end{cases}$$

$$C(y) = \frac{dC(y)}{dy} \rightarrow \text{similar to } f_Y$$

$$\text{Let } \{y_1, y_2, \dots\} \text{ are jump points of } D(y) \quad P(Y = y_k) > 0$$

$$\int_{-\infty}^{\infty} C(y) dy + \sum_{y_k} P(Y = y_k) = 1$$

$$EY = \int_{-\infty}^{\infty} y \cdot C(y) dy + \sum_{y_k} y_k \cdot P(Y = y_k)$$

EX/ Continued

$$a) \text{ Find } P(\frac{1}{4} \leq Y \leq \frac{3}{4}) = ? \quad b) \text{ Find } P(Y \geq \frac{1}{4}) \quad c) \text{ Find } EY$$

Dirac Delta function

$$\text{Recall unit step function: } u(x) = \begin{cases} 1, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

$$\delta(x) = \begin{cases} \infty, & x = 0 \\ 0, & \text{otherwise} \end{cases} \quad \int \delta(x) dx = \frac{d}{dx} u(x)$$

Properties of $\delta(x)$

$$\int_{-\infty}^{\infty} \delta(x) dx = 1 \quad \text{for any } \varepsilon > 0$$

for any $\varepsilon > 0$ and any function $g(x)$ that

is continuous over $(x_0 - \varepsilon, x_0 + \varepsilon)$,

$$\int_{-\infty}^{\infty} g(x) \delta(x - x_0) dx = \int_{x_0 - \varepsilon}^{x_0 + \varepsilon} g(x) \delta(x - x_0) dx = g(x_0)$$

Generalized PDF Discrete

For a discrete r.v. X with range $R_X = \{x_1, x_2, \dots\}$ and PMF $P_X(x_k)$, we define

$$\text{(generalized) PDF as } f_X(x) = \sum_{x_k \in R_X} P_X(x_k) \cdot \delta(x - x_k)$$

Generalized PDF of Mixed R.V.

$$f_X(x) = \sum_k a_k \delta(x - x_k) + g(x), \text{ where } a_k = P(X = x_k) \text{ and } g(x) \geq 0 \text{ does not contain any delta functions}$$

$$\int_{-\infty}^{\infty} f_X(x) dx = \sum_k a_k + \int_{-\infty}^{\infty} g(x) dx = 1$$

$$F_X(x) = \begin{cases} \frac{1}{2} + \frac{1}{2}(1 - e^{-x}), & x \geq 1 \\ \frac{1}{2} - \frac{1}{2}(1 - e^{-x}), & 0 \leq x < 1 \\ 0, & \text{otherwise} \end{cases}$$

$$C(x) \rightarrow F_X(x) = \begin{cases} \frac{1}{4} + \frac{1}{2}(1 - e^{-x}), & x \geq 1 \\ 0, & \text{otherwise} \end{cases}$$

a) what kind of R.V. X ?

$$D(x) \rightarrow \begin{cases} \frac{1}{2}, & x > 1 \\ 0, & \text{otherwise} \end{cases}$$

b) Find the (generalized) PDF of X

c) $P(X > 0.5)$, both using CDF and PDF

d) $E[X] = ?$ Var $X = ?$

solution

1) Graph of $F_X(x)$



mixed

$$2) F_X(x) = \int_{-\infty}^x f_X(t) dt = \int_{-\infty}^x \left(\frac{1}{2} \delta(t) + \frac{1}{2} e^{-t} \right) dt = \frac{1}{2} + \frac{1}{2} (1 - e^{-x})$$

$$3) P(X > 0.5) = \int_{0.5}^{\infty} f_X(x) dx = \int_{0.5}^{\infty} \left(\frac{1}{2} \delta(x) + \frac{1}{2} e^{-x} \right) dx = \frac{1}{2} + \frac{1}{2} (1 - e^{-0.5})$$

joint distributions ch 5-6 read yourself

joint PMF's

EX: Let X and Y be d.r.v.'s joint PMF of X and Y :

$$\begin{array}{c|cc} & Y=0 & Y=1 & \\ \hline X=0 & \frac{1}{8} & \frac{1}{4} & \\ X=1 & \frac{1}{8} & \frac{1}{6} & \end{array} \rightarrow P(X=1, Y=0) = \frac{1}{8}$$

The joint PMF of two discrete r.v.'s X and Y is defined as $P_{XY}(x, y)$

$$R_{XY} = \{(x, y) | P_{XY}(x, y) > 0\} = R_X \times R_Y$$

$$\sum_{x, y} P_{XY}(x, y) = 1 \rightarrow \text{ex } \frac{1}{6} + \frac{1}{4} + \frac{1}{8} + \frac{1}{6} + \frac{1}{6} = 1$$

MARGINAL PMF'S

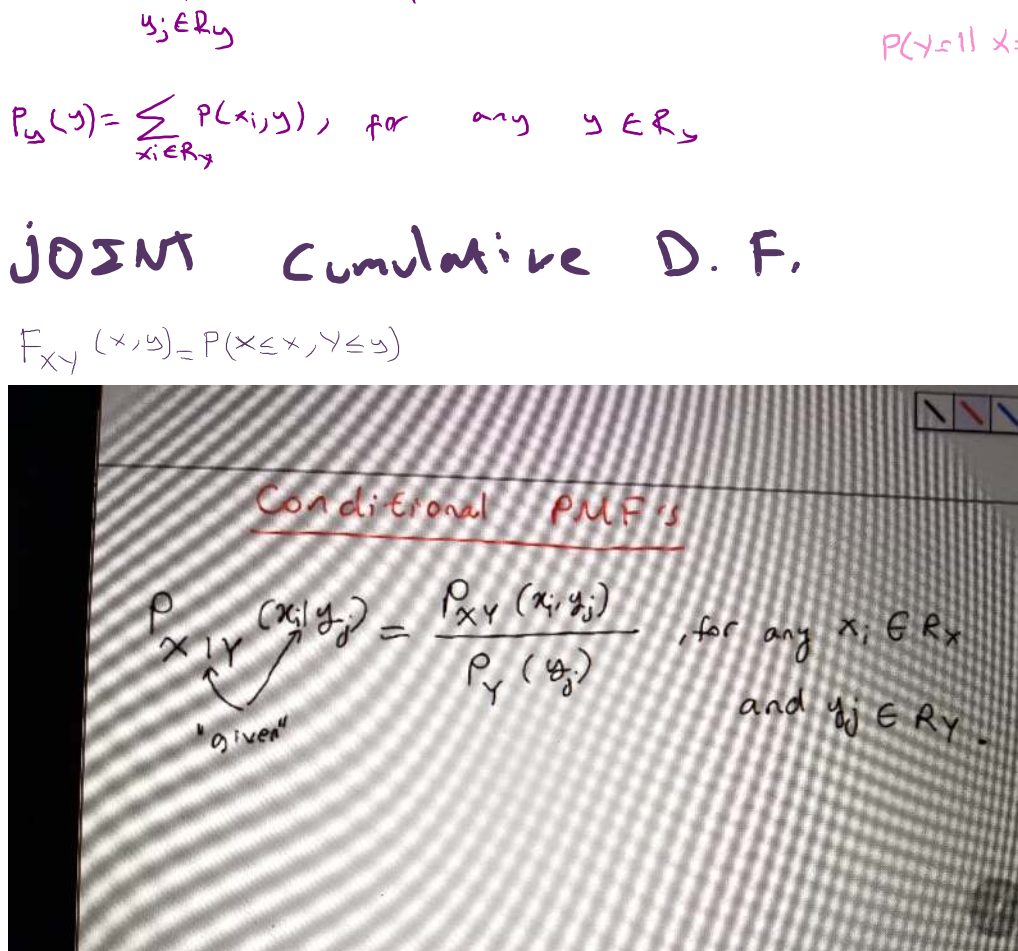
$$P_X(x) = \sum_{y \in R_Y} P_{XY}(x, y), \text{ for any } x \in R_X$$

$$\text{OR continued } P(Y=1 | X=0) = \frac{P(Y=1 \text{ and } X=0)}{P(X=0)} = \frac{P_{XY}(0, 1)}{P_X(0)} = \frac{\frac{1}{4}}{\frac{1}{2}} = \frac{1}{2}$$

$$P_Y(y) = \sum_{x \in R_X} P_{XY}(x, y), \text{ for any } y \in R_Y$$

JOINT cumulative D.F.

$$F_{XY}(x, y) = P(X \leq x, Y \leq y)$$



Conditional expectations

$$E[X | A] = \sum_{x_i \in R_X} x_i \cdot P_{X|A}(x_i), \quad E[X | Y=y_j] = \sum_{x_i \in R_X} x_i \cdot P_{X|Y}(x_i | y_j)$$

Independence of r.v.'s X and Y

Two discrete r.v.'s X and Y are independent if $P_{XY}(x, y) = P_X(x) P_Y(y)$ for all x, y .

Functions of two R.V.'s

$$E[g(X, Y)] = \sum_{x \in R_X} \sum_{y \in R_Y} g(x, y) P_{XY}(x, y)$$

$$E[XY] = \sum_{x=0}^2 \sum_{y=0}^2 xy \cdot P_{XY}(x, y) = 0 \cdot 0 \cdot \frac{1}{6} + 0 \cdot 1 \cdot \frac{1}{4} + 0 \cdot 2 \cdot \frac{1}{8} + 1 \cdot 0 \cdot \frac{1}{8} + 1 \cdot 1 \cdot \frac{1}{6} + 1 \cdot 2 \cdot \frac{1}{6} = \frac{1}{2}$$

Continuous joint R.V.'s

Two R.V. X and Y are jointly continuous if there exists a non-negative $f_{XY}: \mathbb{R}^2 \rightarrow \mathbb{R}$ s.t. for any set $A \in \mathbb{R}^2$, we have $P((X, Y) \in A) = \int_A f_{XY}(x, y) dx dy$

$f_{XY}(x, y)$: joint PDF of X and Y

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x, y) dx dy = 1$$

MARGINAL PDF

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy, \text{ for all } x$$

JOINT CDF

$$F_{XY}(x, y) = P(X \leq x, Y \leq y)$$

$$f_{XY}(x, y) = \frac{\partial^2 F_{XY}(x, y)}{\partial x \partial y}$$

LOTUS C.R.V.

$$E[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f_{XY}(x, y) dx dy$$

Covariance BETWEEN R.V.'s X and Y

(discrete or continuous)

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])] = E[XY] - E[X]E[Y]$$

Correlation R.V.'s X and Y

$$\rho_{XY} = \rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

$\text{Cov}(X, Y) \approx 0$

ρ can take any value $\in \mathbb{R}$

ρ can take values between -1 and 1

$\rightarrow$ high correlation

$\rightarrow$ high negative correlation

$\rightarrow$ no correlation

Properties of Cov

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y)$$

If correlation or Covariance is zero $\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y)$

Ch 5 is ended